

ERROR ANALYSIS TASK

25. A 12-bit uniform quantizer with an input range of ± 2 V gives a quantization step size of 0.002 V. Verify the calculation, identify the error, and compute the correct step size.

Quantization Step Size: Error Analysis

The given quantization step size is 0.002 V. We verify it using the input range and the number of quantization levels.

Given

Parameter	Value
Quantizer resolution (n)	12 bits
Input range	± 2 V
Minimum input	-2 V
Maximum input	+2 V
Given step size	0.002 V

1. Total Input Range

For an input range of ± 2 V, the total range is:

$$\begin{aligned} V_{\text{range}} &= V_{\text{max}} - V_{\text{min}} \\ &= (+2) - (-2) = 4 \text{ V} \end{aligned}$$

2. Number of Quantization Levels

For a 12-bit quantizer, the number of levels is:

$$L = 2^n = 2^{12} = 4096 \text{ levels}$$

3. Correct Quantization Step Size

The quantization step size is calculated as:

$$\begin{aligned} \Delta &= V_{\text{range}} / L \\ \Delta &= 4 / 4096 = 0.0009765625 \text{ V} \\ \Delta &\approx 0.000977 \text{ V} = 0.977 \text{ mV} \end{aligned}$$

4. Verification of the Given Value

The given step size is 0.002 V, whereas the calculated step size is approximately 0.000977 V.

Given $\Delta = 0.002$ V

Correct $\Delta \approx 0.000977$ V

Therefore, the given value 0.002 V is incorrect for a 12-bit quantizer with a ± 2 V input range.

5. Identify the Error

The error is in the calculation of the number of quantization levels or the total input range. A 12-bit quantizer has $2^{12} = 4096$ levels, not approximately 2000 levels.

If 0.002 V were used as the step size, the implied number of intervals would be:

$$4 / 0.002 = 2000 \text{ intervals}$$

This does not match the required 4096 levels for a 12-bit quantizer.

6. Quantization Error

For a uniform quantizer, the maximum quantization error magnitude is half of one step:

$$|e_{\max}| = \Delta/2 = 0.0009765625/2$$

$$|e_{\max}| \approx 0.0004883 \text{ V} = 0.488 \text{ mV}$$

Conclusion

For a 12-bit uniform quantizer with an input range of $\pm 2 \text{ V}$, the total range is 4 V and the number of quantization levels is 4096. Therefore, the correct quantization step size is **$0.0009765625 \text{ V} \approx 0.000977 \text{ V}$ (0.977 mV)**. Hence, the given value of 0.002 V is incorrect. The maximum quantization error is approximately **$\pm 0.0004883 \text{ V}$** .